communication networks, which will determine how drones integrate safely, efficiently, and broadly into the built environment.

9.6.1 How 5G and future networks will enable real-time drone connectivity

The introduction of 5G communication networks significantly enhances the real-time operational capabilities of drones by providing high bandwidth, low latency, and stable connectivity across large and complex environments. Unlike previous networks, where drones frequently faced signal delays or loss of control over long distances, 5G enables near-instantaneous data exchange between UAVs, ground control stations, cloud servers, and other drones in the vicinity. This allows drones to transmit high-resolution video, multispectral sensor data, geolocation information, and flight telemetry continuously and without interruption, which is essential for tasks such as live surveillance, precision agriculture, infrastructure inspection, and emergency response.

Future communication systems particularly 6G, satellite-integrated IoT networks, and edge-cloud computation frameworks are expected to extend this connectivity even further. These networks will allow drones to operate seamlessly in rural wilderness, maritime zones, mountainous regions, and dense urban environments where conventional radio communication struggles. With reliable real-time connectivity, drones can perform beyond visual line of sight (BVLOS) missions safely, coordinate with air traffic control systems, and interact dynamically with other aerial and ground units. Thus, 5G and future networking technologies form the backbone of high-autonomy, intelligent, and scalable drone ecosystems, enabling drones to become integrated, responsive actors within smart city and global mobility infrastructures.